Abdulbaki Sanlan

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Summary

Currently pursuing an PhD. in Aerospace Engineering at Istanbul Technical University and conducting research at the ITU Aerospace Research Center. Experienced in the design and implementation of advanced GNC algorithms for aerospace applications, with a strong focus on developing and applying novel approaches to improve the autonomy, precision, and robustness of aerial systems.

Education

M.Sc. in Aerospace Engineering *2024 – 2025*
Istanbul Technical University, Istanbul, Turkey
GPA: 3.78/4.00

B.Sc. in Aeronautical Engineering *2019 – 2024*
Istanbul Technical University, Istanbul, Turkey
GPA: 3.36/4.00

High School *2016 – 2019*
Ergun Oner Mehmet Oner Anadolu Lisesi, Turkey

Professional Experience

Researcher *Nov 2023 – Present*
ITU Aerospace Research Center, Istanbul, Turkey

- GPS/GNSS Denied Navigation
  - Implemented a **terrain-matching solution** for **6-DOF aircraft state estimation** by fusing **INS data** and **point-cloud measurements** through the **Unscented Kalman Filter (UKF)** and **Particle Filter (PF)**.
  - Developed a **visual navigation-based UAV positioning pipeline** that integrates **Visual Inertial Odometry (VIO)** with **Generative AI(GANs)–enabled domain adaptation** of satellite imagery matching within a **Particle Filter (PF)** framework.
- Anti Skid Breaking System Design
  - Developed an **optimal friction curve estimation** model using **Particle Swarm Optimization (PSO)**, integrating a hybrid approach with a **Sliding Mode Estimator** (SME) for enhanced performance.
  - Conducted research on **effective radius estimation** for systems operating under **measurement-deficient conditions**.

Engineer Intern *Jul 2023 – Sep 2023*
Baykar Teknoloji, Istanbul, Turkey

- Worked on swarm UAV formation flight algorithms based on Artificial Potential Field.

Engineer Intern *Oct 2022 – Mar 2023*
TUSAŞ, Istanbul, Turkey

- Assisted in trim and linearization for a 6-DOF Aircraft model
- Learning visiualization of aircraft model on Simulink/Flight Gear.

Skills

- **Advanced:** Python, MATLAB, LaTeX
- **Intermediate:** ROS, Simulink, Git, C++

Languages

- English (B2, IELTS 7/9)
- Turkish (Native)

Publications

A.B. Sanlan, H. T. Bagci, E. Koyuncu"Particle Filter-Based Localization Using Visual Feature Synchronization in GNSS-Denied Navigation" *2025*
The 25th Integrated Communications, Navigation and Surveillance Conference, Brussels, Belgium.

A.B. Sanlan, F. Erol, M. Abu-Khalaf, E. Koyuncu"Terrain-Aided Navigation Using a Point Cloud Measurement Sensor" *2025*
arXiv e-prints, abs/2510.06470.

Projects

Machine Learning:

- Designed and trained a CycleGAN-based generative model for image-to-image translation, effectively minimizing domain discrepancies across diverse datasets
- Developed and trained a deep learning pipeline for automated species identification using audio recordings, leveraging signal processing and neural network techniques.
- Implemented a grounded image-classification framework built on the Segment Anything Model (SAM), enhancing segmentation precision and object recognition accuracy.

Optimal Control: Finding the trajectory that will take a commercial plane from point A to point B with minimum fuel consumption and minimum time using single shooting, multiple shooting and direct collocation optimal control methods.

Reinforcement Learning: Finding the trajectory that will take a passenger commercial plane from point A to point B with minimum fuel consumption and minimum time with Reinforcement Learning methods, and finding the trajectory that avoids fixed obstacles in a two-dimensional environment and reaches the target.

Dynamic Programming: Modeled one-on-one 3D air combat scenarios and developed sub-optimal dogfighting algorithms using Approximate Dynamic Programming to achieve effective tactical maneuvering under real-time computational constraints.

LQR & LQI Controller: Designed controllers for the F-16 model (Aerobench-vv) using MATLAB/Simulink.

Inverted Pendulum RL Control: Implemented reinforcement learning–based control for an inverted pendulum system.

Volunteer Experience

Software Team Lead on University UAV Team ITU SKY *Sep 2022 - Apr 2024*
ITU, Turkey

- Developing and leading mission planning, using image processing for object detection-tracking and designing custom PID controllers for $TEKNOFEST - 2024$ competition.

Organizational Coordinator *Sep 2022*
SAHA EXPO 2022, Istanbul, Turkey

- Coordinated field setup for national and international exhibitors under ITU Dean's Office.

Participant *Jun 2022 – Sep 2022*
USA Work & Travel Program, Virginia Beach, VA, USA

- Engaged in a three-month cultural exchange program.

Hobbies

Soccer, Computer Games, Ice Skating

References

- Prof. Dr. İbrahim Özkol, Istanbul Technical University, ozkol@itu.edu.tr
- Assoc. Prof. Dr. Emre Koyuncu, Istanbul Technical University, emre.koyuncu@itu.edu.tr
- Dr. Murad Abu-Khalaf, Turkish Aerospace Industries(TF-X Executive Vice President), murad.abu-khalaf@tai.com.tr